Emotion Detection: Using Facial Expressions to Connect to Music

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***Abstract*—Emotion detection has significant applications in human-computer interaction, with facial expressions serving as a critical indicator of emotions. This study explores an approach that leverages image processing techniques to detect emotions and adaptively connect with music. Our system analyzes facial expressions from webcam feeds and links the detected emotions to a music playlist, creating an interactive and responsive experience.**

I. Introduction

Every day, new applications are being created that use image processing to return a result. We see these applications in everyday life such as Snapchat filters or Google's photo AI. In particular, both of these applications use facial recognition to make a determination that meets a particular criteria to return a specific result.

Emotions accompany our lives and affect every aspect. Therefore, emotion detection has great potential for research and exploration. We can use visual information to perceive the emotions of others. Visual information includes facial expressions, eye movements, postures, gestures, gait, etc. Compared with non-visual information, visual information is easier to obtain and more objective. Visual information, such as facial expressions, can be obtained directly by observing others' faces or analyzing images or videos captured by mobile phones, CCTV cameras, or similar devices. Our framework leverages state-of-the-art machine learning techniques such as a pre-trained MobileNetV2 model, which was initially trained on the ImageNet dataset and obtained from the Keras library to achieve high-precision emotion detection, even in challenging environments. This study also includes dataset descriptions, feature extraction, model training and evaluation.

II. Literature Review

A. Current Research in Emotion Detection

Emotion detection is a rapidly growing discipline, with major advances driven by AI approaches such as machine learning and deep learning. Current research in this area focuses on improving the accuracy and usability of emotion recognition systems across multiple modalities, including facial expressions, and physiological signals such as Electroencephalography (EEG) and heart rate data.

Modalities and Techniques: To improve emotion recognition accuracy, researchers are increasingly using multimodal techniques that incorporate data from several sources (e.g., facial expressions, voice and EEG). EEG signals are being used to capture emotional changes, particularly in healthcare and mental health monitoring [1].

Deep Learning Advances: Neural networks, particularly convoluted neural networks (CNNs), are leading the way in facial emotion recognition (FER). CNNs have outperformed classic machine learning approaches (e.g., support vector machines and k-nearest neighbors) due to their improved feature extraction capabilities. [2].

Emotion Detection Applications

- **Public Safety:** Facial Emotion Recognition (FER) systems are increasingly used for security and surveillance, such as detecting stress or anger in high-risk environments [3].
- **Healthcare:** Identifying emotional states can assist in diagnosing mental health conditions and offering personalized treatments.

Challenges & Future Directions: The ability to recognize delicate emotions or distinguish between genuine and synthetic emotions is a major difficulty in emotion detection. Furthermore, researchers are tackling biases in datasets, as most current datasets are insufficiently diverse to capture the full range of human emotions across cultures. Future trends emphasize developing more generalized models and improving the robustness of FER systems, particularly in real-world applications [4].

B. Related Work

We reviewed recent studies in the field of emotion detection and recognition to provide a comprehensive overview of advancements, identify key insights for future research directions, and derive ideas that informed the implementation of our project.

- **A Systematic Literature Review of Modalities, Trends, and Limitations in Emotion Recognition.** [2]
  This review examines recent advances and trends in emotion recognition, sentiment analysis, and affective computing research from 2018 to 2023. The review analyzes 609 articles from 4 databases (IEEE, Science Direct, Springer, and MDPI). It classifies these articles based on the type of data used: unimodal,

multi-physical, multi-physiological, or multi-physical-physiological. Deep learning models, CNNs and RNNs, emerged as the most widely used techniques.

- **New Trends in Emotion Recognition Using Image Analysis by Neural Networks.** [3]
  This paper examined recent advancements in Facial Emotion Recognition (FER) systems that leverage neural networks to analyze images for emotion detection and classification. It demonstrated the effectiveness of CNNs and transfer learning for image-based FER. However, high computational requirements and dataset diversity remain a challenge.
- **eMotions: A Large-Scale Dataset for Emotion Recognition in Short Videos.** [6]
  This large-scale dataset of 27,996 short videos provides a benchmark for multimodal emotion recognition. Its baseline model, AV-CPNet, employs cross-modal fusion techniques for improved accuracy, making it valuable for audiovisual emotion detection tasks and suited for handling incomplete or imbalanced emotional data.

III. Methodology

A. Datasets Description

In our project, we tested several datasets and computer vision models as follows:

1. **OAHEGA Dataset:**
The OAHEGA Emotion Recognition Dataset, is a collection of RGB images featuring cropped faces, each labeled with one of six distinct emotions: Happy, Angry, Sad, Neutral, Surprise and Ahegao. The term ”Ahegao” refers to a facial expression characterized by exaggerated features, often associated with certain internet subcultures. These images were collected by scrapping social networks such as Facebook and Instagram. The dataset was published in April 2021 by Vinnytsia National Technical University. Researchers have utilized the OAHEGA dataset in various studies. For example, it has been combined with the FER-2013 dataset to train Convolutional Neural Network (CNN) models for facial emotion recognition tasks. [8]
We excluded the Ahegao face expression from our project to improve the accuracy because it is important to note that expression may influence the dataset’s applicability depending on the context.

2. **AffectNet Dataset:**
This is a comprehensive, large-scale dataset widely used for facial expression recognition and emotion analysis in computer vision. The dataset has seven basic emotions: Happy, Sad, Angry, Fearful, Disgusted, Surprised and Neutral. Each image in the dataset is labeled with continuous values for valence (positive vs. negative emotion) and arousal (intensity of the emotion). It contains more than 1 million facial images collected from the internet by querying three major search engines using 1250 emotion-related keywords in six different languages. About 450,000 of the images are manually labeled, and the other 550,000 are automatically labeled images. [9] [10] [11]

Fig. 1. A Sample of OAHEGA Dataset.

It is important to mention some of the challenges in the AffectNet dataset.

1) Ambiguity in Labeling: some facial expressions are inherently ambiguous, leading to challenges in consistent labeling.
2) Imbalance: certain emotions, such as ”Neutral” and ”Happy”. are overrepresented compared to less common emotions like ”Fear” or ”Disgust”

3. **FER-2013 Dataset:**
The Facial Expression Recognition 2013 (FER-2013) dataset is a widely utilized resource in the field of facial emotion recognition. It was introduced during the ICML 2013 challenge in Representation Learning and has become a standard benchmark for evaluating facial expression recognition algorithms. [12]
The dataset contains 35,887 grayscale images, each with a resolution of 48 x 48 pixels. These images are centred on faces displaying seven emotional expressions: Angry, Disgust, Fear, Happy, Sad, Surprise and Neutral. The images were collected using the Google image search API, and the dataset is split into three sets: [13]

- Training Set: 28,709 images
- Public Test Set: 3,589 images
- Private Test Set: 3,589 images

The FER-2013 dataset is commonly used for training and benchmarking emotion recognition models due to its diversity and the relatively large number of labeled samples. However, its grayscale nature and low image resolution may limit model generalizability for real-world applications.

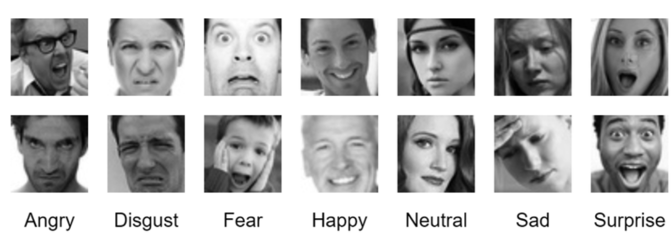

Fig. 2. A Sample of FER-2013 Dataset.

4. **FERPlus Dataset:**

The FERPlus dataset is an enhanced version of the original Facial Expression Recognition (FER) dataset, developed to improve the quality and accuracy of emotion recognition tasks. In FERPlus, each image has been labeled by 10 crowd-sourced taggers, providing better quality ground truth for still image emotion than the original FER labels.

Some of this dataset features are each image in FERPlus has been labeled by 10 crowd-sourced taggers, enabling researchers to estimate an emotion probability distribution per face. The dataset includes eight emotion categories: Neutral, Happiness, Surprise, Sadness, Anger, Disgust, Fear, and Contempt. FERPlus maintains the same number of rows and order as the original FER dataset, facilitating straightforward integration and comparison between the two. [14]

5. **RAF-DB Dataset:**

The Real-world Affective Faces Database (RAF-DB) is a dataset for facial expression. This version contains more than 15,000 facial images tagged with basic or compound expressions by 40 independent taggers. Images in this dataset are of great variability in subjects' age, gender and ethnicity, head poses, lighting conditions, occlusions, (e.g. glasses, facial hair or self-occlusion), post-processing operations (e.g. various filters and special effects), etc. The dataset has seven basic emotion categories: Happy, Sad, Angry, Fear, Disgust, Surprise, and Neutral. It also has unique compound emotions such as: Happily surprised, sadly fearful, etc., for nuanced affect recognition. Each image is manually annotated by multiple human labelers to ensure reliability. One of the strengths of this dataset is its diversity, compound emotions and real-world scenarios where it introduces challenging conditions like partial occlusions, varying lighting, and diverse facial characteristics, making it ideal for robust model training. [15]

In our project, we applied and tested the previous datasets to train and fine-tune deep learning models for real-time facial emotion detection as shown in the subsequent sections.

B. Models Architecture

We tested the MobileNetV2, ResNet50, and EfficientNetB0 models, all pre-trained on ImageNet and fine-tuned them using OAHEGA, AffectNet, FER-2013 and FERPlus datasets. For real-time processing, we integrated OpenCV to handle webcam input, face detection, and image preprocessing. Detected faces were resized to match each model's input requirements, ensuring efficient and accurate inference for real-time emotion detection.

1. **ResNet50:** We tried The ResNet50 architecture on our selected datasets. This model is known for its deep residual learning framework, was pre-trained on ImageNet and fine-tuned on the AffectNet dataset. ResNet50's skip connections mitigate vanishing gradient issues, making it effective for extracting complex features from facial images. This model was tested for its ability to generalize across diverse datasets.

2. **EfficientNet:** EfficientNet, designed with a compound scaling approach, was employed for its efficiency in balancing accuracy and computational cost. We utilized the EfficientNetB0 variant, pre-trained on ImageNet, and fine-tuned it on our selected datasets. This model demonstrated promising results in scenarios requiring resource-constrained deployment.

3. **MobileNetV2:** MobileNetV2 was prioritized for real-time applications due to its lightweight design and efficient depth-wise separable convolutions. The model was pre-trained on ImageNet and fine-tuned using both the AffectNet and FER-2013 datasets. Its low computational requirements enabled smooth real-time emotion detection using webcam feeds.

4. **ImageNet Pre-Trained Models:** To establish a baseline, we evaluated pre-trained models without fine-tuning specific emotion datasets. These models provided insights into the out-of-the-box capabilities of architectures such as MobileNetV2, ResNet50, and EfficientNet for emotion recognition.

IV. Problem Statement

The purpose of this project is to build a real-time emotion detection system that takes in a webcam feed to analyze facial expressions and play music according to the detected emotional state. With this, we have a responsive environment in which the "mood of the house" could be recorded and improved by playing music actively. This project had to solve the challenges of real-time performance, handling multiple faces, summing up emotions into a collective mood, and properly integrating into a web-based platform. The project was implemented using Python as the programming language, along with frameworks and libraries such as TensorFlow and Keras for deep learning model deployment, Flask for web-based application hosting, OpenCV for webcam input and face detection, and NumPy for data processing.

A. Challenges and Solutions

1. **Real-Time Performance:** A significant challenge was ensuring that emotion detection works efficiently with low latency for smooth real-time interactivity for a smooth user experience. We accomplish this by using a pre-trained MobileNetV2 model from the TensorFlow and Keras frameworks, which is particularly well-suited for mobile and edge applications because of its efficient architecture. We integrated the model with OpenCV for processing the frames from the webcam in real-time, while providing an acceptable frame rate to maintain a smooth user experience.

2. **Multi-Face Detection:** Initially, our model was designed to detect emotions from a single face. We expanded this functionality to handle multiple individuals simultaneously by iterating over all detected faces in each frame. Using OpenCV's Haar Cascade, we detected multiple faces and processed each one to predict emotions independently. This required additional computational resources, but we optimized the process to ensure minimal lag by only updating the mood of the house once per frame.

3. **Aggregating Emotional States:** To determine the "mood of the house" when multiple faces were detected, we required a method for adding the emotion of the individuals. We used a majority voting mechanism to identify the collective mood based on the most frequent emotion present among all faces in a frame. In cases where emotions were tied, we selected

randomly certain emotions to stay responsive. We followed this approach to get the overall mood that best represents the emotional state of everyone.

4. **Integration with Web Application:** We established the system using a Flask web app to make the system easier to use. Using JavaScript, we were able to send information back and forth between the backend where the model was making predictions and the frontend where the feed from the webcam was being displayed and music was played. The integration was in the form of streaming the webcam feed to the browser and dynamically changing the music according to the emotions detected so that the user would have a cohesive experience.

5. **Model Accuracy Improvement:** When training models on Google Colab and utilizing a GPU, we tuned hyperparameters (such as number of layers trainable from the base model), trained more epochs, and incorporated some data augmentation techniques (rotation, flipping, zooming, etc.) to make the model more tolerant to different expressions in the face.

V. Methods

The system utilized either pre-trained MobileNetV2, ResNet50, or EfficientNet-B0 models, which were initially trained on the ImageNet dataset and obtained from the Keras library. These models were then fine-tuned with either of the OAHEGA, AffectNet, or FER-2013 datasets to detect emotions in real-time during several experiments. The overall architecture integrated facial detection, emotion recognition, and dynamic music playback through a web application interface.

A. Preprocessing

Using Python and OpenCV the video feed from the webcam was preprocessed in real-time for efficient emotion detection in three stages:

1. **Face Detection:** After the webcam feed was converted to grayscale from RGB, Haar Cascade classifiers were used for face detection in each frame. This helped the system from computational load and focused its processing power on later regions of interest.

2. **Face Cropping and Resizing:** Detected faces were cropped and resized for uniformity. In the case of MobileNetV2 and ResNet50, they were cropped and resized to a 48 x 48 pixel resolution to be in the same input size required by these models. As in the case of EfficientNet they were resized to a 224 x 224 pixel resolution to match the input shape of the model.

3. **Normalization:** Finally, NumPy pixel values were normalized between 0 and 1 to make them appropriate for neural networks, and this helped in accurate predictions.

B. Model

In several experiments, the emotion detection model was built using either MobileNetV2, ResNet50, or EfficientNet from the TensorFlow and Keras libraries, which were pre-trained on the ImageNet dataset and fine-tuned with either the OAHEGA, AffectNet or the FER-2013 datasets.

In this process, first, we prepared the fine-tuning dataset. This involved several techniques in the case of the OAHEGA dataset for instance, rows with the label 'Ahegao' were removed from the dataset for data cleaning, after cleaning the data we preprocessed the labels by converting categorical labels to integer indices and One-hot encoded them for multi-class classification. Next, we split the dataset into 90% for training/validation, and 10% for testing. After this step, we preprocessed the images first by decoding them into tensors, then resizing them into the corresponding base model input shape, and finally normalizing the pixel values to [0,1]. In the next step, we built the model architecture by using a base pre-trained model of either MobileNetV2, ResNet50, or EfficientNet-B0 on ImageNet. Then we customize it by adding custom layers, such as GlobalAveragePooling2D for feature reduction, Dropout to prevent overfitting, and a Dense output layer for classification (The number of classes was based on the fine-tuning dataset). After this, we compile/configure the model with the loss of categorical cross entropy, Adam optimizer with various learning rates and metrics of accuracy. After we created the model we used 5-fold cross-validation to train and evaluate the model on each fold. At the same time, we recorded metrics like validation loss and accuracy for each fold. In this 5-fold cross-validation step, we applied early stopping to prevent overfitting and learning rate reduction on plateau callbacks for training. After this, we evaluated the final model on the unseen 10% test dataset.

Finally, when the model achieved a satisfactory result, it was saved as a .keras file and metrics, classification reports, and system information were saved into a JSON file for recording each experiment. Later on, the model was loaded into a Flask application to process incoming webcam frames, predict emotions, and return results for real-time display and music playback.

The model serves the following purposes in the context of the web application:

1. **Input Shape:** It accepts an input of shape (48, 48, 3) in the case of MobileNetV2 and ResNet50, and (224, 224, 3) in the case of EfficientNet, representing the resized and normalized RGB face images.

2. **Output:** It Produces a probability distribution over the seven emotion classes: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral. In the case of the FER-2013 dataset and five classes: Angry, Happy, Neutral, Sad, and Surprise, in the case of OAHEGA.

3. **Integration:** The model was loaded into the Flask application to process incoming webcam frames, predict emotions, and return results for real-time display and music playback.

C. Aggregation of Emotions

When several faces were detected, the system makes a decision as the "mood of the house" using a majority voting approach:

1. **Emotion Detection for Each Face:** The model predicted the emotion for each detected face, and the results were stored in a list.

2. **Majority Voting:** The most common emotion is determined using Python's `counter.most_common()` method. This majority emotion was then used to decide the music that would be played.

3. **Handling Ties:** In the case of ties, the system randomly selects one to maintain consistency in the played music.

D. Real-Time Music Playback on Spotify

The system actively plays music from Spotify based on the collective mood of the detected faces:

1. **Music Selection:** A detected mood is matched to a particular Spotify playlist URL. For example: "Happy" is matched with the Upbeat playlist while "Sad" corresponds to Melancholic Playlist. when the mood of the house changes, the playlist URL changes accordingly.

2. **Spotify Integration:** First, the users at the front end log into their Spotify accounts through OAuth2 authentication. The app then obtains a valid Spotify access token for API interaction. Using this the app checks if there are active Spotify devices at the user's side, once found, it plays the selected playlist on the device found.

3. **Real-Time Updates:** After some time, which depends on how much time we set before updating the playlist, the app detects the mood of the house and updates the Spotify playlist for active user experience.

E. Loss Function

The categorical cross-entropy loss function was used for training the model. This loss function is defined as:

$$\text{Loss} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{1}$$

Where:

- C: is the number of classes (seven emotions).
- y_i: is the true label for class i (1 if the sample belongs to class i, otherwise 0).
- $\hat{y}_i$: is the predicted probability that the sample belongs to class i.

We applied categorical cross-entropy loss as the difference between the predicted emotion probabilities and the true labels. Adam optimizer was used during training to minimize this loss function. By minimizing this loss the model was trained to predict the correct probability distributions for all the emotion classes, which led to a better classification performance.

F. Evaluation Metrics

For performance assessment of the emotion detection model, Four evaluation metrics were adopted:

1. **Accuracy:** a measure that is defined as the ratio of the number of accurately predicted emotions to the number of samples. It was used to adjust both train and validation steps.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

2. **Confusion Matrix:** A confusion matrix was used to give a detailed breakdown of the model's performance of each emotion class, showing which emotions were often mistaken for one another.

3. **Overall Model Performance:** The difference between training and validation accuracy was monitored to prevent the model from overfitting and generalize well to new data.

4. **Precision, Recall, and F1-Score:** These were computed for each emotion class to provide an overall evaluation of the model's classification capabilities, especially with the issue of class imbalance.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

VI. RESULTS

We evaluated the performance of MobileNet, ResNet50, and EfficientNet models on FER-2013, AffectNet, RAF-DB and OAHEGA datasets using key metrics such as accuracy, precision, recall, and F1-score. Additionally, we measured the computational performance in terms of inference time and memory usage under CPU and GPU setups. All of our experiments are listed in Table 1 below along with the datasets and their metrics. According to the table and experiments, our winning model was the ResNet50 with OAHEGA dataset. We achieved an accuracy of **84.28%**.

In another experiment we used the OAHEGA dataset with the MobileNetV2 model. The images were preprocessed to a resolution of 224 x 224 and the labels were one-hot encoded for multi-class classification. For the training configuration, we used Adam optimizer with learning rate = $1e-4$ and batch size = 32. This training was conducted in Google Colab with GPU acceleration. The precision of this model with the dataset ranged from 63% to 100%, with higher precision for dominant classes. The recall value ranged from 53% to 81%. And the F1-score ranged from 62% to 78%, reflecting balanced performance across classes. The confusion matrix of the experiment displayed imbalances and some misclassifications, especially in underrepresented classes and the visualization demonstrated stronger predictions for more frequent classes.

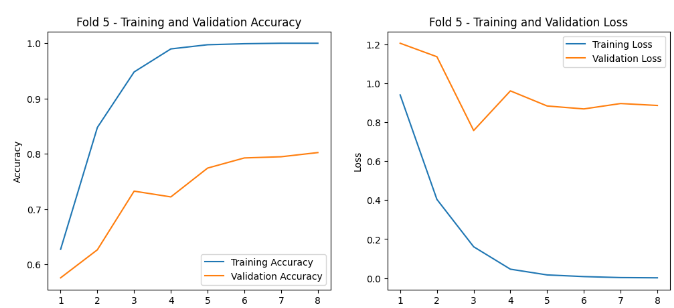

Fig. 3. Training and Validation Accuracy/Loss - ResNet50 with OAHEGA

TABLE I
PERFORMANCE METRICS FOR MODELS ON DIFFERENT DATASETS

Model	Dataset	Accuracy (%)	F1 Score	Precision	Recall	Loss	AUC
ResNet50	**OAHEGA**	**84.28**	**0.8169**	**0.8220**	**0.8155**	**0.7321**	**0.9640**
	RAF-DB	78.32	0.6681	0.7245	0.6646	0.6480	0.946
MobileNetV2	OAHEGA	73.25	0.793	0.8316	0.7873	0.7574	0.9493
	RAF-DB	76.96	0.7767	0.7797	0.779	0.691	0.9392
	FERPlus	60.81	0.185	0.1817	0.1957	1.733	0.5042
EfficientNet	AffectNet	62.76	0.7402	0.7821	0.6942	0.7146	0.8819

VII. DISCUSSION

The main objective of this project was to design a real-time emotion detection application with music adaptation capabilities. The application can classify facial expressions and dynamically play a music playlist based on the emotional state of the person. The ResNet50 model trained on the OAHEGA dataset achieved the highest accuracy of 84.28% with an F1-score of 81.6%, outperforming MobileNetV2 on the same dataset. The integration of real-time webcam input and the Spotify API successfully demonstrated the system's practical application in real-world scenarios. Also, the incorporation of diverse datasets like OAHEGA, RAF-DB, and AffectNet allowed for robust model evaluation across varied contexts. Despite the system's success, certain limitations were observed when the system struggled with overlapping emotions and multi-face scenarios. This project showcases the potential of integrating emotion detection with adaptive music systems for applications in smart homes and entertainment. For example, it can be integrated into a front door camera system in a smart home and based on the facial expression of a person coming, a playlist will be played inside the home.

VIII. CONCLUSION & FUTURE WORK

This project successfully demonstrated a real-time emotion detection system with the ability to analyze facial expressions and adapt a music playlist from Spotify according to the emotional state. The system achieved notable performance metrics. However, some of the challenges we faced in our development included dataset biases and class imbalances.

Future work may focus on increasing the accuracy by expanding the training datasets to include more diverse facial expressions, age groups and ethnicity. Another area of future work is to optimize the models by exploring advanced neural networks such as transformers. We hoped that our system would build the foundation for further exploration of emotion detection and adaptive technologies in personal and professional domains.

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- We are grateful to the developers of TensorFlow, Keras, OpenCV, and Flask for creating robust tools that significantly eased our development process.

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